

Optical Switching for AI Factories

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Abstract: As advances in AI drive productivity, the need for expansive GPU clusters rises. Meeting this demand calls for unprecedented bandwidth, improved energy efficiency and robust network fabrics. We examine optical switching as an architectural response.

1. Introduction

Over the past few years, the rise of generative AI has transformed conventional data centers into AI factories [1]: industrial-scale engines that produce intelligence instead of physical goods. Training trillion-parameter models and serving them to billions of users has pushed compute demand to grow by multiples every year, turning GPU-accelerated infrastructure from a niche capability into the core of modern cloud and enterprise architectures. The global build-out of AI factories now involves clusters with tens of thousands of accelerators, supported by roadmaps that point to continued large-scale expansion of high-end GPU capacity over the coming years.

Each new GPU generation substantially increases both compute throughput and memory/IO bandwidth. The NVIDIA A100 Tensor Core GPU offers up to 80 GB of HBM2e with over 2 TB/s of memory bandwidth and up to 600 GB/s of NVLink bandwidth per GPU when fully connected in an HGX A100 server. The H100 generation moves to HBM3 with around 3 TB/s of memory bandwidth and fourth-generation NVLink, delivering 900 GB/s of total NVLink bandwidth while the associated NVSwitch fabric increases total switch throughput to 13.6 Tbit/s from 7.2 Tbit/s in the prior generation.

Blackwell-class systems [2], extend this trajectory: Blackwell GPU is paired with HBM3e delivering up to approximately 8 TB/s of memory bandwidth and fifth-generation NVLink providing 1.8 TB/s of bidirectional bandwidth per GPU. In the GB200 NVL72 design, up to 72 Blackwell GPUs are connected in a single NVLink domain, and NVLink Switch technology can provide an aggregate of 130 TB/s of low-latency GPU communication. These improvements mean that per-GPU local bandwidth has increased by roughly a factor of four from A100 to Blackwell, while the total NVLink bandwidth available within a single tightly integrated system has grown by more than an order of magnitude, concentrating unprecedented communication capacity into each new GPU system.

As these rack-scale systems become the basic building blocks, AI factories also need to scale out across many racks. GB200-class racks can be composed into modular scalable units and then extended to deployments of thousands of racks, using a multi-tier InfiniBand or Ethernet fabric to deliver near full-bisection bandwidth across the cluster. Recent work on Spectrum-X Ethernet and scale-across technologies [3], further extends this idea beyond a single room or building, allowing multiple data centers to operate as one logical AI factory while maintaining predictable latency and throughput for multi-trillion-parameter training workloads.

As these AI factories scale to hundreds of thousands of accelerators, the bottleneck is no longer just floating-point throughput on a single GPU but the ability to move vast volumes of data with low latency and high reliability across the entire cluster. In this context, networking is effectively becoming the new backplane of the AI system, and electrical interconnects alone are hitting their practical limits in bandwidth density, reach, and power efficiency. Alongside new GPU architectures, there is a parallel investment in high-performance networking and silicon photonics to enable AI factories where the network and the compute are co-designed as a single system.

In this talk, the focus will be on optical switching for AI factories. The discussion will center on how optical switching can assist on delivering the bandwidth, energy efficiency, and reliability needed to scale AI clusters, and how the Optical Circuit Switching (OCS) technology could influence the way AI factories are designed at rack, row, and data center scale.

2. Related Work

Recent growth in AI workloads has renewed interest in optical circuit-switching technologies. The immense bandwidth requirements and the structured communication patterns of these workloads have motivated several new research directions. In this context, [4] explores the integration of OCS into torus-based architectures, while [5] investigates systems that leverage the predictable traffic patterns common in deep learning applications to optimize OCS usage. Looking further ahead, the authors in [6] introduce a quantitative framework for evaluating OCS networks in large-scale AI systems exploring optimal reconfiguration strategies. Several other works are targeting AI workloads in recent years.

3. OCS in AI factories

In our previous work [7], we investigated how optical switching can extend network programmability down to the physical layer, identifying two key use-cases for OCSs: enhancing resilience to failures and enabling topologies that can adapt to the communication patterns of deep-learning workloads. A small-scale prototype was deployed to validate these ideas, demonstrating both dynamic resilience mechanisms and application-aware topology adjustments, along with initial software components to control a reconfigurable Layer 1 infrastructure.

The current presentation builds on those concepts, as illustrated in Figure 1. Scale-out OCSs interface with the electrical switching fabric - either between layers or in place of entire layers - supporting dynamic topology adjustment to match the traffic characteristics of AI applications or to improve reliability by introducing spare paths. These concepts were explored further in prior work.

At the server layer, DPU-attached OCSs provide additional programmability across the cluster. Large-scale deployments require extensive cabling and significant bring-up time; OCS-enabled programmability can accelerate this process by correcting mis-cabled connections and simplifying system assembly. When combined with scale-out OCSs, they enable direct accelerator-to-accelerator links and can help prevent server disconnections during component failures.

Scale-across OCSs allow flexible interconnection of buildings or geographically distributed locations. These systems can take advantage of higher optical loss budgets and the widespread use of wavelength-division multiplexing (WDM) transceivers typical of longer-reach deployments.

As noted earlier, most of the system's bandwidth is concentrated at the scale-up layer, making this the most challenging - but also the most impactful - place to introduce OCS technology. Deploying OCSs here can flexibly extend high-bandwidth domains across racks without incurring the full cost of additional transceivers and electrical switches. They can also help create larger pooled resource domains, improving the availability and resilience of high-bandwidth interconnects.

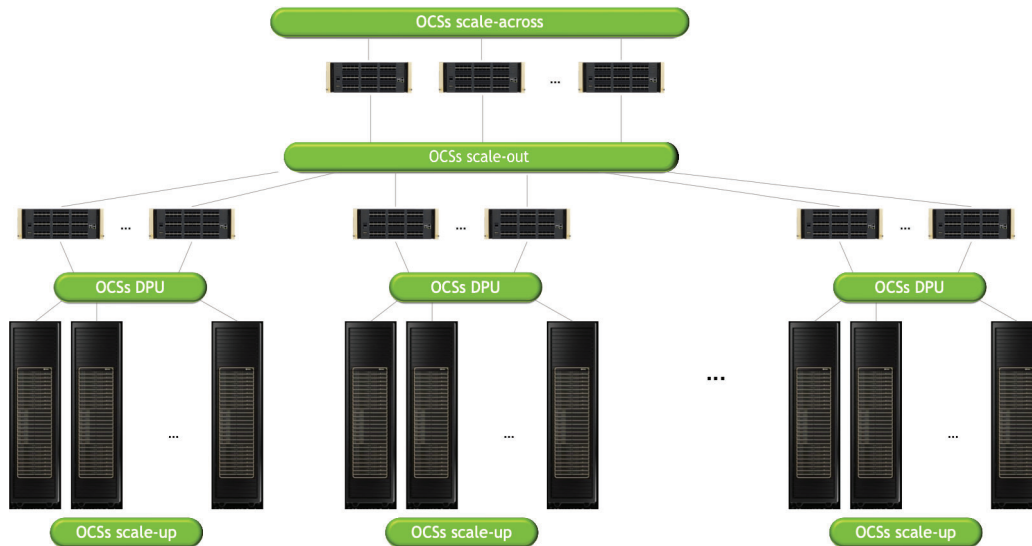


Figure 1: Potential OCS insertion points in the AI Factory

4. OCS technology

Free-space Optical Circuit Switches, like [8-9], are now being used at scale in modern data centers and AI clusters [10]. The ability to dynamically reconfigure network topologies becomes increasingly valuable as these systems continue to grow in size and complexity. However, despite the growing momentum behind OCS adoption, the technology remains relatively bespoke. Continued progress is necessary to further lower insertion loss, minimize costs and space requirements, and validate overall system reliability.

One of the primary technical hurdles facing OCS technology is insertion loss, which consumes a significant portion of the optical link budget. While extended-budget transceivers can help compensate for these losses, they do so at the expense of increased cost and power consumption. Therefore, ensuring that OCSs are compatible with future transceivers and the overall link budget remains a critical consideration.

Wavelength Division Multiplexing (WDM) offers a means to mitigate cost and space concerns by enabling multiple lanes to share a single fiber pair, thus increasing the effective radix of the OCS. However, relying on a greater number of wavelengths can reduce fiber fanout and introduce additional insertion loss. Bidirectional transmission is another option for improving port utilization, but in certain configurations, it can limit any-to-any connectivity. As the scale of OCS deployments increases, costs are expected to decrease. Integrated "solid-state" OCS platforms [11-13] promise further reductions, but their higher insertion loss may necessitate on-chip amplification, which adds complexity, power consumption and optical impairments to the overall link design.

The large and intricate nature of AI systems demands that OCS components achieve exceptionally high reliability, since a single fault can affect many network ports. Achieving this requires comprehensive reliability modeling and careful system design that considers optics, electronics, power, control logic, and redundancy. Miniaturization of OCS devices and the design of "dense racks" will also be essential, as the number of required connections grows with each new generation of AI systems.

5. Conclusion

Optical circuit switching is a promising technology for scaling AI factories, offering the potential for high bandwidth, improved energy efficiency, and flexible topology reconfiguration. It is likely to become increasingly useful as clusters grow and networking demands continue to rise. To fully realize its benefits, we must overcome current technological drawbacks - such as insertion loss, cost, and density - and complement these improvements with thoughtful codesign.

6. References

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